Demystifying the Black Box: A Comparative Analysis of Interpretability, Sequence Modeling, and Decision Theory in ECG Arrhythmia Classification

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ABSTRACT
Evaluating ECG classifiers solely on accuracy obscures a critical trade-off: model capacity versus diagnostic transparency. Using 100,707 beats from the MIT-BIH Arrhythmia Database under a strict patient-disjoint protocol, we compare three architectures spanning the transparency spectrum: a decision tree, a feed-forward neural network (MLP), and an attention-based recurrent model. While macro-F1 performance scales monotonically with capacity ($0.183 \rightarrow 0.288 \rightarrow 0.345$), explanation quality degrades inversely. Using a uniform attribution interface, we demonstrate that as capacity rises, explanation plausibility (clinical localization) falls from $0.87 \rightarrow 0.50 \rightarrow 0.29$, and faithfulness (predictive reliance) drops directionally from $0.68 \rightarrow 0.50 \rightarrow 0.24$. We further show that cost-sensitive thresholding dramatically improves MLP rare-class recall ($0.14 \rightarrow 0.84$) by exploiting fine-grained probabilities, a mechanism unavailable to the tree. Ultimately, this quantitative cross-check confirms that higher diagnostic capacity directly compromises verifiable algorithmic trust.

1 Problem Statement

Automated arrhythmia detection from surface ECGs is a natural candidate for machine learning: the signal is abundant and carries the diagnostic morphology (QRS complex, T-wave) that cardiologists already analyze. However, deploying a classifier as a clinical decision-support tool introduces a severe tension. A tool that cannot explain *why* it flagged a sequence - using clinically actionable features - cannot be trusted or audited. This creates a trade-off that raw accuracy metrics obscure: the highest-capacity architectures are inherently the least transparent.

This report makes this tension measurable. We pose three core questions: (1) Under a rigorous patient-disjoint evaluation, how do three models spanning the transparency spectrum compare on imbalance-aware classification metrics? (2) For each model, is its explanation both *plausible* (localized on relevant clinical regions) and *faithful* (actually relied upon for the prediction)? (3) How do these two properties co-vary with model capacity? Our central contribution is a controlled comparison using a uniform attribution interface, quantifying the exact cost of adopting high-capacity sequence models over interpretable baselines.

2 Introduction

Cardiac arrhythmias are among the most common findings in ambulatory ECG monitoring, and their timely, correct classification directly informs clinical triage: a missed ventricular ectopic beat carries a very different cost than a missed normal beat mislabeled as abnormal. Over the last decade, the machine learning literature on heartbeat classification has followed a familiar arc - from hand-engineered features paired with shallow classifiers, to convolutional and recurrent architectures that consume the raw waveform directly and outperform the former on aggregate accuracy (1). This progression has quietly traded away something clinicians actually need: a legible account of *why* a given beat was flagged. A cardiologist auditing a decision-support tool can read a threshold on QRS width; they cannot read a 128-dimensional hidden state.

This report is a controlled experiment along exactly that capacity axis. We hold the data, the labels, the patient-disjoint evaluation protocol, and the evaluation metrics fixed, and vary only the architecture: a depth-bounded decision tree over clinically-grounded features, a feed-forward network over the raw waveform, and a bidirectional recurrent network with additive attention

over the same raw waveform. Since every architecture's explanation is reduced to the same object - a length-187 per-timestep importance vector, scored by the same plausibility and faithfulness metrics - the three models are comparable not just on *how well* they classify, but on *how honestly* their stated reasons match their actual computation.

The remainder of the report proceeds as follows. Section 3 (Background) situates the project against the heartbeat-classification and explainable-AI literatures it draws on. Section 4 (Methodology) describes the MIT-BIH data pipeline, the patient-disjoint split, and all three model families, followed by the uniform attribution contract used to score every explanation. Section 5 (Results and Analysis) reports headline classification metrics for each model, the plausibility/faithfulness cross-check, the cost-sensitive decision-theoretic re-thresholding, and the accuracy-inflation gap between honest and leaky evaluation protocols. Section 6 discusses limitations, and Section 7 concludes.

3 Background

Heartbeat classification. The De Chazal et al. (2004) protocol is the de facto standard for AAMI-class heartbeat classification on MIT-BIH: it defines the inter-patient DS1/DS2 record split we adopt, the mapping from raw MIT-BIH annotation symbols to the five AAMI classes, and a feature set built from RR-interval timing and QRS morphology (1). Later work replaced these hand-engineered features with representations learned directly from the raw waveform - convolutional networks, and subsequently recurrent and attention-based sequence models - which consistently improve aggregate classification metrics but discard the direct feature-to-decision mapping that made the earlier, shallower classifiers auditable. Our three architectures are chosen specifically to span this progression rather than to chase a single state-of-the-art number: a decision tree at the interpretable end, a feed-forward network as the mid-capacity opaque baseline, and a recurrent attention model at the high-capacity end.

Post-hoc and native explanations. Where a model has no built-in explanation, we rely on established post-hoc attribution methods: gradient-based saliency and gradient$\times$input, Integrated Gradients, which accumulates gradients along a path from a baseline input to the true input to satisfy a completeness axiom (4), and SHAP, which approximates Shapley-value feature attributions via sampling (5). Where a model has a built-in mechanism that resembles an explanation - here, the softmax attention weights of the recurrent model, following the additive attention formulation of Bahdanau et al. (6) - we do not take that mechanism at face value. Jain and Wallace (2019) show that attention weights can vary substantially without changing a model's prediction, undermining the assumption that attention indicates what the model relied on (2); Wiegreffe and Pinter (2019) offer a partial rebuttal but agree that attention should be corroborated rather than assumed faithful (3). We adopt the conservative reading: attention is a *candidate* explanation, always cross-checked against an independent post-hoc method.

Plausibility, faithfulness, and decision theory. The distinction between an explanation that *looks* right (plausibility) and one the model *actually used* (faithfulness) is well established in the interpretability literature, and it is the organizing axis of this report's central contribution. We measure plausibility as the fraction of an attribution's importance mass that falls inside clinically annotated QRS/T-wave windows, and faithfulness via occlusion - masking the most important timesteps and measuring the resulting drop in predicted probability. Separately, because MIT-BIH is severely class-imbalanced, we borrow the standard cost-sensitive decision-theoretic framing: rather than reporting only argmax accuracy, we re-threshold each model's predicted probabilities to minimize expected cost under an explicit cost matrix that penalizes missed abnormal beats more heavily than false alarms, exposing a second, orthogonal way in which model capacity matters - not just for accuracy, but for how much room a downstream decision rule has to work with.

4 Methodology

4.1 Data

All data is sourced from the MIT-BIH Arrhythmia Database (7), which was sourced from PhysioNet via the `wfdb` toolkit. The database contains 48 half-hour, two-channel ambulatory ECG recordings sampled at 360 Hz, featuring beat-by-beat annotations verified by cardiologists. The modified-limb-lead-II (MLII) channel is used exclusively, as it is consistently available and best captures the QRS morphology required for clinical feature extraction.

Rather than using pre-segmented datasets, the raw records were dynamically segmented in-house. This methodological choice preserves the mapping from each beat to its source patient, which is a strict prerequisite for constructing the patient-disjoint evaluation split. Following the ANSI/AAMI EC57 recommendations, four paced records (102, 104, 107, 217) were entirely excluded. Their morphology is dominated by pacemaker artifacts rather than intrinsic cardiac activity, which would otherwise confound the arrhythmia models. The remaining 44 records gave a total of 100,707 beats.

Class Distribution. Beats are mapped to the five standard AAMI morphological classes. As detailed in Table 1, the dataset is severely imbalanced. Normal (N) beats dominate the corpus (89.5%), whereas fusion (F) and unclassifiable (Q) beats represent less

than 1% combined. The Q class is effectively unlearnable under an inter-patient protocol and is reported solely for completeness. Due to this extreme skew, aggregate accuracy is discarded as a headline metric; all models are evaluated using macro precision, recall, F1-score, AUROC, and AUPRC to properly reflect minority-class performance.

Table 1: AAMI Class Distribution across the filtered 44 MIT-BIH records.

Class	Meaning	Count	Share
N	Normal / bundle-branch / escape	90,101	89.5%
S	Supraventricular ectopic	2,781	2.8%
V	Ventricular ectopic	7,008	7.0%
F	Fusion (ventricular + normal)	802	0.8%
Q	Unknown / paced / unclassifiable	15	0.0%

4.2 Data Cleaning and Processing

The raw ECG records were processed into a single, deterministically aligned artifact consisting of normalized per-beat arrays. This pipeline ensures total reproducibility from the raw PhysioNet download through a standardized sequence of steps. First, during annotation filtering, only genuine beat annotations were retained, as MIT-BIH annotations include rhythm and signal-quality markers alongside beat labels. These retained annotations were mapped to the five AAMI classes following the standard De Chazal et al. (2004) protocol (Table 2), while unrecognized symbols were treated as non-beat markers and discarded.

Table 2: Mapping of MIT-BIH annotation symbols to standard AAMI classes.

AAMI Class	MIT-BIH Symbols
N	`N L R e j`
S	`A a J S`
V	`V E`
F	`F`
Q	`/ f Q`

Following annotation filtering, each retained beat was segmented and extracted as a half-open window spanning indices $[R - 90,\ R + 96)$, the 90 samples preceding the R-peak, the R-peak itself, and the 95 following it, giving 186 native samples (≈ 0.52 seconds at 360 Hz). Any edge beats lacking sufficient samples were discarded. Crucially, this R-peak centering ensures that the QRS-complex and T-wave regions are aligned at fixed offsets across all beats, providing the unambiguous spatial foundation required for the downstream localization (plausibility) cross-checks. These native 186-sample windows were then resampled to 187 samples (Fourier method, `scipy.signal.resample`), producing the standard fixed-length vector used throughout the ECG literature. Under this fixed geometry, the R-peak maps to index $\mathrm{round}(90 \times 187/186) = 90$ in every beat; the sub-sample residual (≈ 0.48) is negligible relative to the $\pm$15-sample QRS and $+$20–60-sample T-wave windows anchored to it. To remove inter-recording baseline drift and gain variance, each beat was subsequently independently min-max normalized to a $[0, 1]$ range. Consequently, downstream amplitude-derived features are measured relative to each beat's dynamic range rather than in absolute millivolts.

Next, inter-beat metadata was extracted by computing the temporal intervals from each beat's R-peak to the preceding (RR_{prev}) and subsequent (RR_{next}) R-peaks in seconds. This RR timing carries critical discriminative signal for supraventricular ectopy (S). For the first and last beats in a record, missing intervals mirror their available neighbor to prevent zero-padding artifacts from distorting the decision tree splits. Finally, to prevent data leakage, the 44 records were partitioned into training and testing sets strictly at the patient (record) level, mirroring the De Chazal protocol. The training set (DS1) comprises 22 records: 101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, and 230. The test set (DS2) comprises the remaining 22 records: 100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, and 234. This inter-patient split ensures models are evaluated on genuinely unseen patient morphologies rather than memorized waveforms. This constraint is enforced algorithmically via the project's test suite, which asserts zero shared patients between DS1 and DS2 prior to evaluation, and all headline metrics report performance exclusively on the DS2 test set. For the leakage comparison in Section 5.11 we additionally define an intra-patient split: all beats are pooled across the 44 records and partitioned 80/20 into train and test by stratified random sampling on the AAMI class label (seed 42), so beats from the same patient appear on both sides.

Models evaluated under this protocol are retrained from scratch on the leaky training set rather than merely re-scored. Note that this split also supplies 80,565 training beats against the inter-patient split's 51,009, so a portion of any apparent inflation reflects the larger training set rather than leakage alone.

4.3 Decision Tree Model

At the interpretable end of the capacity spectrum, we fit a depth-bounded (depth-6) decision tree using scikit-learn. Unlike the two neural models, which consume the raw 187-sample waveform, the tree is deliberately restricted to six derived, clinically-grounded features: the two RR intervals (RR_{prev}, RR_{next}), QRS width and amplitude, T-wave amplitude, and ST-segment level. A tree that splits on raw sample index 90 is not clinically interpretable - "sample 90" means nothing to a cardiologist - whereas a split on `qrs_amplitude` or `rr_prev` reads directly in the vocabulary a clinician already uses. This is the entire point of the tree: its capacity is deliberately bounded so that its full decision logic can be printed, read, and audited end to end. The depth bound of 6 is an interpretability constraint fixed *a priori* rather than a tuned hyperparameter: it is the largest depth at which the complete root-to-leaf path set (57 leaves) still fits in a single readable listing. No validation sweep over depth was performed, and the tree reported here should accordingly be read as the most expressive fully-printable tree rather than the best-performing one.

The tree is trained with `class_weight="balanced"` in addition to explicit per-sample weights passed to `fit`. We flag this as a probable methodological error rather than a design choice: scikit-learn *multiplies* `class_weight` by `sample_weight`, so the correction for the imbalance documented in Table 1 is applied twice and the effective minority-class up-weighting is approximately squared. This pushes the decision boundary far harder toward the minority classes than a single correction would, and is the most likely cause of both the tree's collapsed majority-class recall (N recall 0.333, Section 5.1) and its anomalous inflation gap (Section 5.11). All tree results in this report are reported as-trained under this double correction; re-fitting with a single correction is the obvious next step and is noted in Section 6.

4.4 Feed-forward Neural Network (MLP)

The MLP occupies the mid-capacity point between the transparent tree and the high-capacity attention model. It consumes the same raw 187-sample beat waveform as the attention model (rather than the tree's six derived features), passed through a stack of fully-connected Linear/ReLU/Dropout layers to a five-way softmax output. Critically, the MLP has no built-in explanation mechanism of any kind - no decision path, no attention weights - so every one of its explanations in Section 5 is entirely post-hoc (Integrated Gradients and SHAP). This makes the MLP the cleanest test of what capacity alone costs an explanation: it sits strictly between the tree and the attention model in representational capacity, is explained by the same post-hoc methods used on the attention model, and lets us isolate the effect of *capacity* on explanation quality from the effect of *architecture family* (recurrent vs. feed-forward).

Concretely, the MLP has two hidden layers of width 256 and 128, each Linear layer followed by ReLU and dropout ($p = 0.3$), then a linear head to the five-way output; its input is the raw 187-sample waveform only, with no RR features. It is trained for 30 epochs with Adam (learning rate 10^{-3}, batch size 256) under a per-sample-weighted cross-entropy loss, using the same training loop and seeded batch-shuffling as the attention model. One protocol difference must be flagged rather than glossed over: the MLP uses full inverse-frequency class weighting, whereas the final attention model (Config C) uses the milder square-root weighting. Because the tuning arc in Section 5.6 shows that full inverse-frequency weighting substantially depresses the attention model's own macro-F1 in the closest matched comparison available (0.218 with full weighting versus 0.292 unweighted, both without RR features), the two neural models are *not* matched on imbalance strategy, and the classification gap between them cannot be read as a pure architecture effect (see the caveat in Section 5.10). Finally, dropout and Adam introduce run-to-run non-determinism, so all MLP numbers in this report are drawn from a single canonical trained checkpoint, re-evaluated identically for every metric, explanation, and decision-theoretic analysis.

4.5 Attention Sequence Model

To evaluate high-capacity representations, we implemented a recurrent encoder with additive attention. Rather than consuming hand-derived features, this model processes each beat as an ordered 187-step sequence.

Architecture: The encoder is a single-layer bidirectional GRU (hidden size 128) that produces a 256-dimensional hidden state per timestep. An additive (Bahdanau) attention layer scores each hidden state, outputting a class-agnostic softmax attention map over the 187 timesteps. These weights pool the hidden states into a single context vector and serve as the model's native temporal explanation. The context vector is passed through a dropout layer ($p = 0.3$) to a linear classifier head. Crucially, RR intervals

(RR_{prev}, RR_{next}) are appended directly at the classifier head rather than fed into the sequence recurrence. This ensures the attention map remains strictly over the 187-sample waveform, preserving structural comparability with the baseline models.

Training and Imbalance Strategy: The model was trained for 30 epochs using the Adam optimizer (learning rate 10^{-3}, batch size 256) with a per-sample-weighted cross-entropy loss. Due to the extreme class imbalance, four configurations were evaluated, crossing weighting strength against the RR feature inclusion. The final selected model utilizes a mild (square-root) class weighting combined with the RR feature, as this uniquely improved headline macro-F1 performance while recovering recall for the rare supraventricular (S) class.

4.6 Explainability Cross-Check

To rigorously compare explanations across architectures, we defined a uniform attribution contract where every explanation method outputs a normalized 187-step importance array, enabling a single evaluation code path for all models. Each architecture is explained using techniques tailored to its structure. For the sequence model, we extract the native softmax map to evaluate attention weights. For the neural models, we apply gradient-based methods, specifically saliency, gradient$\times$input, and Integrated Gradients, which utilizes a 50-step Riemann approximation from a zero baseline. Additionally, SHAP is implemented as a model-agnostic permutation explainer applied to the output probabilities of all models. Finally, for decision trees, we trace the native root-to-leaf paths and distribute the impurity-decrease weights uniformly over the relevant clinical feature windows.

A measurement caveat exists for the sequence model: appending the RR features at the classifier head forces post-hoc explainers to evaluate the waveform against a zero-baseline RR. This introduces an out-of-distribution artifact. Consequently, the capacity-versus-interpretability trade-off is analyzed as a within-model directional trend rather than a strictly absolute cross-model ranking.

The generated explanations are evaluated across three dimensions: plausibility, faithfulness, and corroboration. Plausibility, or localization, measures whether an explanation aligns with human clinical intuition. Because the beats are R-peak centered at index 90, the clinical regions occupy fixed windows: the QRS complex (R-peak ± 15 samples) and the T-wave (+20 to +60 samples). Localization is scored as the fraction of normalized attribution mass falling within these specific windows, with a uniform importance map establishing a baseline score of approximately 0.385.

However, plausibility alone does not guarantee model reliance, which necessitates measuring faithfulness. Faithfulness is evaluated via occlusion, where the top 20% most important timesteps are masked to a zero baseline and the resulting drop in target-class probability is recorded. A larger probability drop indicates a more faithful explanation. For the neural models, this masked waveform is passed directly back into the network. For the decision tree, the occlusion protocol must bridge the gap between temporal timesteps and derived clinical features. When the tree's top 20% most important timesteps are masked to zero on the raw waveform, this mutilated signal is re-featurized on the fly. The morphological features (QRS width, QRS amplitude, T-wave amplitude, and ST-segment level) are re-derived directly from the occluded signal before being passed back to the tree. This guarantees that the tree's faithfulness score evaluates its true reliance on the underlying waveform exactly as the neural models do, producing a strictly comparable probability drop.

4.7 Decision-Theoretic Thresholding

Argmax classification minimizes the raw error count, treating every misclassification as equally costly. In a clinical setting this is the wrong objective: missing a genuine abnormal beat (a false negative on S, V, or F) is materially more dangerous than raising a false alarm on a Normal beat. We therefore re-threshold every model's predicted class probabilities using an explicit cost matrix in which missing an abnormal beat costs five times as much as a false alarm, selecting for each beat the class that minimizes expected cost rather than the class with the highest raw probability (`bayes_decision`). This is applied identically to the decision tree and the MLP, using each model's own `predict_proba` output, so that the comparison isolates a single variable: how much usable structure each model's probability output actually contains. As Section 5 shows, this reframing exposes a capacity effect that is orthogonal to raw accuracy - a model can be re-thresholded usefully only if its probability estimates are fine-grained enough to move.

5 Results and Analysis

All reported metrics are derived from a single canonical run on the MIT-BIH dataset using the patient-disjoint inter-patient split (DS1 train, DS2 test; 49,698 test beats). Explanation scores are computed over a fixed random sample ($n = 30$) of DS2 test beats (see §4.6). Unless otherwise specified (e.g., §5.6), the attention model reflects the final selected configuration: square-root

class weighting with RR features appended at the classifier head. All code, configuration files, and the evaluation harness used to produce these numbers are available at `https://github.com/sanasarin/ecg-arrhythmia-xai`.

5.1 Decision Tree: Classification Performance

The decision tree is the least capable classifier of the three by every headline metric (Table 3): macro-F1 of 0.183, AUROC of 0.671, and AUPRC of 0.261. Its per-class breakdown (Table 4) tells a very different story from the neural models', however. The doubled `class_weight="balanced"` correction avoids concentrating recall on the majority class. It spreads recall relatively evenly across all five classes (0.33 / 0.12 / 0.22 / 0.34 / 0.43).

Achieving this balance destroys minority-class precision. Normal-beat recall collapses to 0.333, meaning the tree flags roughly two-thirds of genuinely normal beats as abnormal. Flooding the rare classes with these false positives tanks their precision (e.g., S-class precision drops to 0.034). This exposes the direct cost of correcting for imbalance with a low-capacity model: there simply are not enough splits available to carve out the minority classes without sacrificing the majority class. Figures 1 and 2 show where those errors land and how precision and recall separate by class.

Table 3: Decision Tree Headline Metrics (Inter-Patient Split).

Metric	Value
Accuracy (context only)	0.318
Macro-F1	**0.183**
Macro AUROC	0.671
Macro AUPRC	0.261

Table 4: Decision Tree Per-Class Performance.

Class	Precision	Recall	F1
N	0.995	0.333	0.499
S	0.034	0.118	0.053
V	0.674	0.220	0.332
F	0.016	0.335	0.031
Q	0.000	0.429	0.000

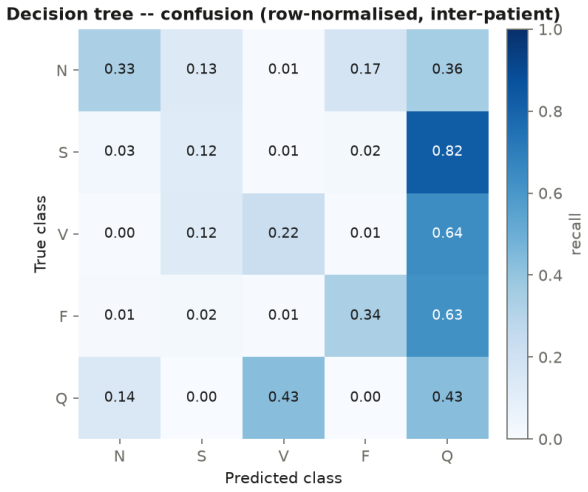


Figure 1: Decision tree confusion matrix (row-normalised). The near-uniform, off-diagonal mass illustrates the tree's low overall discriminative capacity - rather than a single systematic failure mode, errors are dispersed across nearly every class pair.

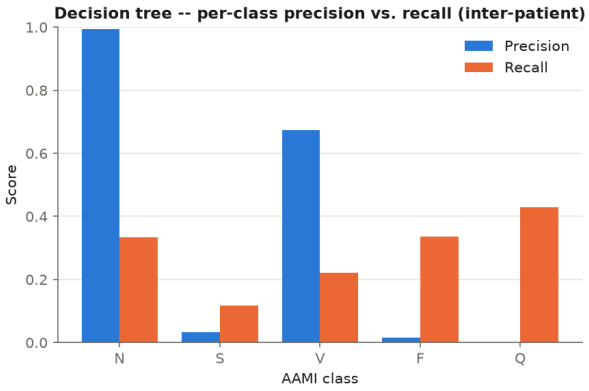


Figure 2: Decision tree per-class precision vs. recall. Precision is high only for the classes the tree can cleanly separate on the six clinical features (N, V); recall is spread thin across all five classes by the balanced class weighting.

5.2 Decision Tree: A Readable Decision Path

The entire motivation for restricting the tree to derived clinical features (Section 4.3) is that its decision path can be read in the vocabulary a cardiologist already uses. The top of the fitted tree (57 leaves total, depth 6) begins:

```
|--- qrs_amplitude <= 0.82
|   |--- rr_next <= 0.49
|   |   |--- rr_prev <= 0.55
|   |   |   |--- t_wave_amplitude <= 0.18
|   |   |   |   |--- qrs_amplitude <= 0.59
|   |   |   |   |   |--- qrs_width <= 15.50  -> class N
|   |   |   |   |   |--- qrs_width >  15.50  -> class V
|   |   |   |   |--- qrs_amplitude >  0.59
|   |   |   |   |   |--- rr_prev <= 0.33     -> class V
|   |   |   |   |   |--- rr_prev >  0.33     -> class S
...
```

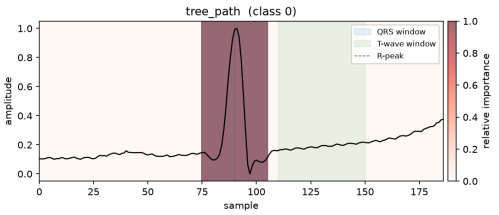


Figure 3: Tree-path importance overlay on a representative DS2 test beat. Since the tree's features are, by construction, drawn from the QRS/T-wave/ST windows, its localization score is trivially 1.0 - a structural property of a feature-engineered explanation rather than evidence of a more discerning model.

Every split reads as `qrs_amplitude <= 0.82`, not `feature_2 <= 0.82` - a condition a cardiologist recognizes on sight, and the direct payoff of feeding the tree derived features instead of raw sample indices. This is also why the tree's native explanation (the root-to-leaf path, converted into a per-timestep importance map over the same QRS/T-wave/ST windows used by the localization scorer) achieves a plausibility score of 1.000 *by construction*: it can only ever point at the clinical regions it was built from. Figure 3 shows this native explanation rendered over an actual test beat.

5.3 Feed-forward MLP: Classification Performance

The MLP sits at the mid-capacity point of the spectrum, and its headline metrics fall between the tree and the attention model on nearly every axis (Table 5): macro-F1 of 0.288, AUROC of 0.747, and AUPRC of 0.353 - each clearly above the tree's, and below the attention model's on macro-F1 and AUPRC, though the MLP in fact edges it out on threshold-free AUROC (0.747 vs. 0.739). Its per-class profile (Table 6) already resembles the attention model's more than the tree's: recall concentrates on the two learnable classes (N: 0.709, V: 0.801) while the rare classes (S, F, Q) remain weak, the same qualitative pattern that intensifies further at the attention model's higher capacity (Section 5.5). Figures 4 and 5 show the confusion structure and the per-class precision/recall trade-off.

Table 5: MLP Headline Metrics (Inter-Patient Split).

Metric	**Value**
Accuracy (context only)	0.690
Macro-F1	**0.288**
Macro AUROC	0.747
Macro AUPRC	0.353

Table 6: MLP Per-Class Performance.

Class	**Precision**	**Recall**	**F1**
N	0.942	0.709	0.809
S	0.054	0.139	0.078
V	0.381	0.801	0.517
F	0.020	0.245	0.036
Q	0.000	0.000	0.000

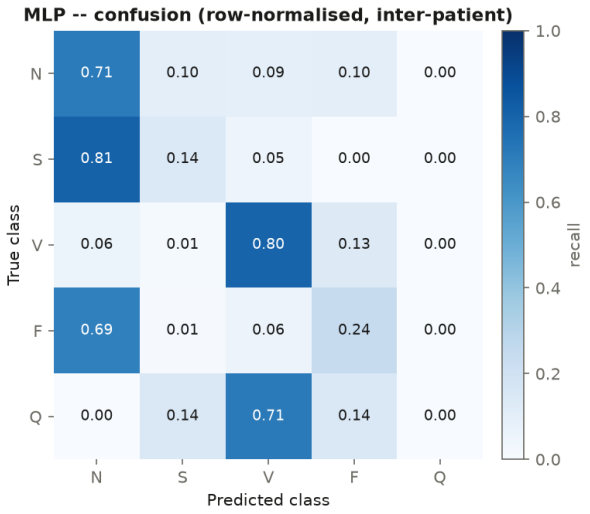


Figure 4: MLP confusion matrix (row-normalised). N and V recall is comparatively strong, but S, F, and Q are largely absorbed into the majority classes - the same failure mode that appears, and intensifies, in the attention model (Figure 7).

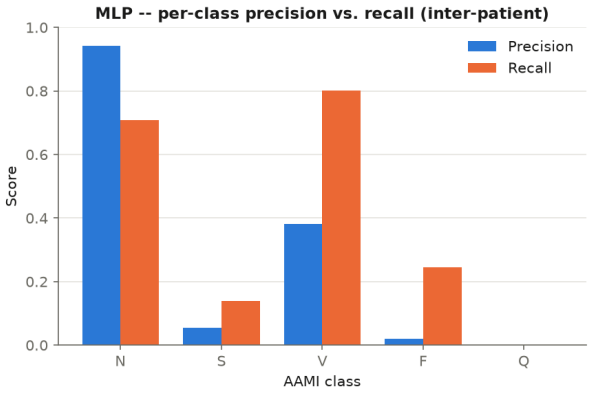


Figure 5: MLP per-class precision vs. recall. V recall is high but at low precision, mirroring the imbalance penalty seen in the higher-capacity attention model.

5.4 Feed-forward MLP: Post-Hoc Explainability

The MLP has no native explanation mechanism, so both of its attribution methods are entirely post-hoc: Integrated Gradients and SHAP, applied directly to the raw 187-sample waveform with no RR out-of-distribution caveat (unlike the attention model). Table 7 reports plausibility (localization) and faithfulness for both methods.

Table 7: Plausibility and Faithfulness, MLP (post-hoc only).

Method	Localization (Plausibility)	Faithfulness
Integrated Gradients	0.524	0.484
SHAP	0.500	0.502

Both post-hoc methods land comfortably above the 0.385 uniform-importance baseline and, unlike the attention model's Config C (Table 11), are computed fully in-distribution - there is no zero-baseline RR artifact to caveat. Faithfulness is likewise meaningfully higher than the attention model's (Table 12; 0.484–0.502 vs. 0.193–0.202), meaning that occluding the MLP's top-20%-important timesteps changes its output far more than the same intervention does for the attention model. Figure 6 shows representative overlays for both methods on the same test beat.

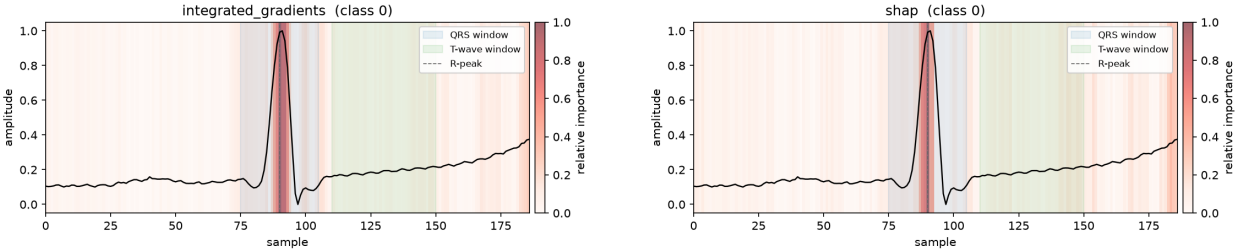


Figure 6: MLP post-hoc importance overlays on a representative DS2 test beat: Integrated Gradients (left) and SHAP (right). Both concentrate more of their importance mass on the shaded QRS/T-wave windows than the attention model's corresponding maps (Figure 10).

This is the cleanest evidence in the report that explanation degradation tracks *capacity*, not merely the native-vs-post-hoc distinction: the MLP's post-hoc Integrated Gradients map is both more plausible (0.524 vs. 0.386) and more faithful (0.484 vs. 0.202) than the attention model's Integrated Gradients map, computed by the identical method on the identical class of post-hoc explainer (Section 5.10, Table 16).

5.5 Attention Model: Classification Performance

Because Normal (N) beats constitute approximately 90% of the corpus, headline performance is evaluated using macro precision, recall, F1-score, and the threshold-free ranking metrics AUROC and AUPRC (Table 8). Aggregate accuracy is reported strictly for context.

Table 8: Attention Model Headline Metrics (Inter-Patient Split).

Metric	Value
Accuracy (context only)	0.777
Macro-F1	**0.345**
Macro AUROC	0.739
Macro AUPRC	0.375

Table 9: Attention Model Per-Class Performance.

Class	Precision	Recall	F1
N	0.951	0.804	0.871
S	0.089	0.118	0.101
V	0.596	0.880	0.711
F	0.000	0.003	0.000
Q	0.025	0.143	0.043

The sequence model concentrates its predictive capacity on the two most learnable classes (Table 9): it recalls 80.4% of Normal beats and 88.0% of ventricular ectopic (V) beats, the most common clinically abnormal class. However, the three rare classes remain challenging. Supraventricular (S) ectopy is detected weakly (recall 0.118 at very low precision), fusion (F) beats are essentially missed (recall 0.003), and Q is statistically negligible. This reflects a genuine limitation of single-lead morphology rather than a tuning artifact; the raw waveform combined with RR timing simply lacks sufficient discriminative signal to isolate F and Q classes.

The row-normalised confusion matrix (Figure 7) exposes the structure of these errors: S and F beats are largely absorbed into the N class, while Q is misclassified as V; Figure 8 gives the corresponding precision/recall split. A single aggregate accuracy score would obfuscate this critical failure mode entirely.

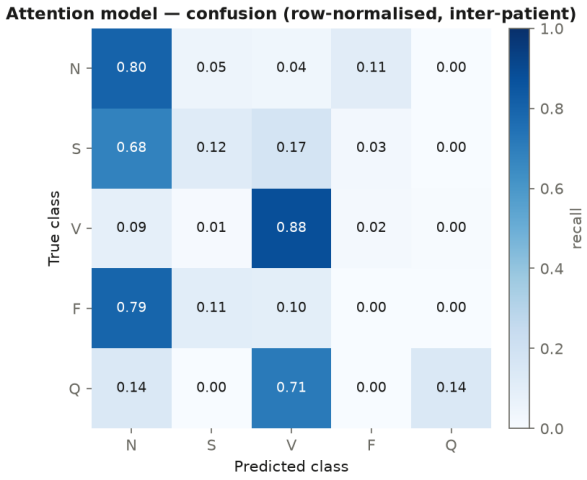


Figure 7: Attention confusion matrix (row-normalised). The diagonal represents per-class recall, exposing the structural absorption of S and F into N, and Q into V.

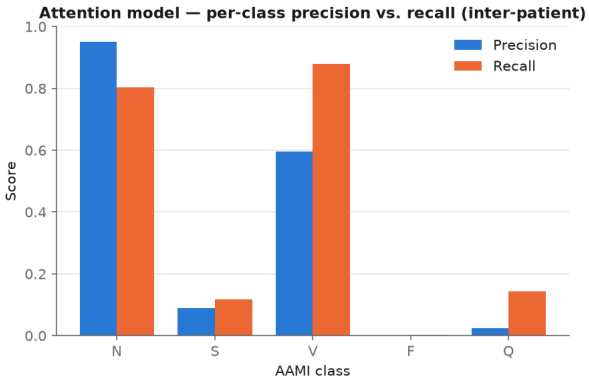


Figure 8: Attention per-class precision vs. recall. The plot illustrates the severe imbalance penalty: V is recalled effectively but at modest precision, while S, F, and Q remain highly constrained.

5.6 The Tuning Arc: Class Weighting and RR Features

The class-weighting strength and RR feature inclusion were selected empirically to navigate the dataset's extreme imbalance. Four configurations were evaluated on the inter-patient split. Performance was measured via macro-F1, while S-class recall was tracked independently as the minority class most sensitive to weighting adjustments; Table 10 reports all four.

Table 10: Attention Model Tuning Arc across weighting strategies and RR inclusion.

Config	Weighting	RR	Macro-F1	S Recall
Baseline	Full inverse-frequency	No	0.218	0.130
A	None	No	0.292	0.050
B	None	Yes	0.341	0.010
C (Final)	**Mild ($\sqrt{x}$)**	**Yes**	**0.345**	**0.118**

Full inverse-frequency weighting over-corrected: aggressively pursuing rare classes destroyed Normal-beat recall, collapsing macro-F1 to 0.218. Removing weighting entirely (Config A) recovered overall performance to 0.292 but virtually abandoned the S class (recall 0.05). Appending RR features without weighting (Config B) lifted macro-F1 to 0.341 by sharpening majority-class separation, but S-recall dropped to an unacceptable 0.01. Only **Config C (mild square-root weighting with RR)** successfully lifted the headline metrics (macro-F1 0.345, best AUPRC 0.375) while simultaneously recovering meaningful S-class recall (0.118); Figure 9 plots the progression across all four configurations.

Crucially, this tuning arc exposes a stark classification-versus-interpretability tension *within* the attention family itself:

Table 11: Interpretability degradation across the tuning configurations.

Config	Macro-F1	AUPRC	Localization	Corroboration (IG / SHAP)
A (No RR)	0.292	0.352	**0.695** (in-dist)	+0.24 / +0.24
B (RR)	0.341	0.372	0.912 (OOD baseline)	+0.22 / +0.32
C (RR, Final)	0.345	0.375	0.257 (OOD baseline)	+0.33 / +0.34

Config A is objectively the cleanest interpretable sequence model. Without RR features, its attention map evaluates fully in-distribution and localizes at 0.695 (well above the 0.385 uniform baseline). The addition of RR features introduces the out-of-distribution (OOD) zero-baseline caveat, but the contrast between Config B (0.912) and Config C (0.257) reveals exactly how that measurement artifact behaves. Config B abandons the rare, timing-dependent S-class almost entirely (recall 0.010); because it only successfully classifies morphologically-distinct majority beats, its attention remains hyper-focused on the QRS window, driving localization to 0.912. Config C, however, uses class weights to force the model to detect those rare S-beats. Because the model must now rely heavily on the RR features to make these minority-class decisions, the explainer's OOD zero-baseline on those exact features causes the waveform attention to completely diffuse (0.257). Ultimately, Config A represents the best strictly-valid explanation, whereas Config C represents the only viable clinical classifier - and the gap between them highlights the severe measurement cost of appending out-of-sequence features.

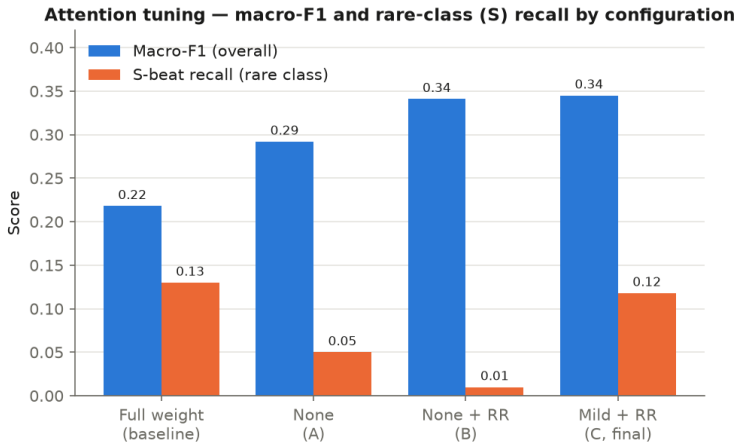


Figure 9: Tuning progression comparing Macro-F1 and S-class recall across the four configurations.

5.7 Explainability: Plausibility and Faithfulness

Plausibility and faithfulness are distinct constructs and are measured independently. For the final high-capacity model (Config C), all three attribution methods score poorly on both axes:

Table 12: Plausibility and Faithfulness across attribution methods (Attention Config C).

Method	Localization (Plausibility)	Faithfulness
Attention weights	0.257	0.193
Integrated Gradients	0.386	0.202
SHAP	0.289	0.241

Two phenomena are evident. First, the native attention map's localization (0.257) falls strictly *below* the 0.385 uninformative uniform baseline, and Integrated Gradients (IG) barely matches it (0.386). The model's predictive importance is not concentrated on the clinical QRS/T-wave windows. While partially attributable to the RR zero-baseline measurement artifact, this also reflects genuine capacity-driven diffusion.

Second, and entirely independent of the baseline caveat, **faithfulness remains uniformly low (0.19–0.24)** across all methods. Occluding the top 20% most-important timesteps marginally alters the target prediction. The high-capacity model's reasoning is highly diffuse across the entire temporal sequence rather than dependent on localized, occludable clinical features. Even when an explanation appears visually plausible, the model does not mathematically rely upon it.

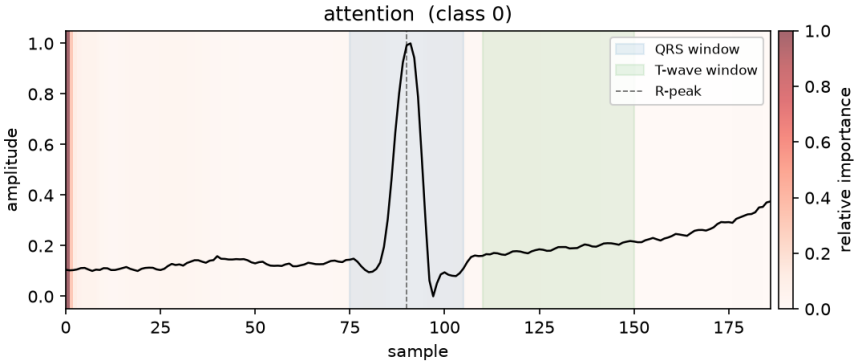


Figure 10: Attention importance overlay on a representative correctly-classified DS2 beat. The highlighted background represents the attention map, contrasted against the shaded QRS and T-wave windows defined by the localization scorer.

5.8 Corroboration: Validating Attention

Following established literature on the unreliability of native attention, the attention weights were explicitly corroborated against independent attribution methods (IG and SHAP). Table 13 reports the per-beat Spearman rank-agreement over the $n = 30$ sample.

Table 13: Spearman rank-agreement validating Attention against independent methods.

vs. Attention	Mean	Median	Std	n	Degenerate
Integrated Gradients	**+0.330**	0.366	0.212	30	0
SHAP	**+0.340**	0.395	0.266	30	0

Agreement is **modestly positive**. The attention weights carry non-random signal (both means and medians are positive with zero degenerate flat maps), but they lack strong confirmation. A correlation of roughly +0.33 leaves the majority of the ranking variance unexplained. This weak corroboration validates the methodological requirement to treat attention strictly as a *candidate* explanation rather than ground-truth reasoning.

5.9 Cost-Sensitive Decision Thresholding

Re-thresholding by expected cost (Section 4.7) rather than raw argmax reveals a capacity effect entirely separate from classification accuracy. On the **MLP**, the cost matrix changes 37.5% of all decisions, because the model's roughly 48,978 effectively-distinct probability vectors across 49,698 test beats give it fine-grained room to move. The effect on rare-class recall is dramatic (Table 14): S-class recall rises from 0.139 to 0.841, at the expected cost of overall accuracy (0.690 $\rightarrow$ 0.376) - exactly the trade a clinical deployment would want, favoring sensitivity to abnormal beats over raw correctness.

The **Decision Tree** tells the opposite story (Table 15). The same cost matrix changes only 12.2% of decisions, because a depth-6 tree's `predict_proba` takes just **36 distinct probability vectors** across its 57 leaves, so cost-sensitive re-thresholding can only flip decisions at the coarse, whole-leaf level. Recall is essentially unchanged for every class except N, which drops slightly (0.333 $\rightarrow$ 0.196) as a handful of leaves flip their assigned class.

Table 14: MLP recall under argmax vs. cost-sensitive (`bayes_decision`) thresholding.

Class	Recall (argmax)	Recall (bayes)
N	0.709	0.325
S	0.139	**0.841**
V	0.801	0.819
F	0.245	0.291
Q	0.000	0.000

Table 15: Decision tree recall under argmax vs. cost-sensitive (`bayes_decision`) thresholding.

Class	Recall (argmax)	Recall (bayes)
N	0.333	0.196
S	0.118	0.120
V	0.220	0.220
F	0.335	0.335
Q	0.429	0.429

Read together, these two tables make the point of Section 4.7 concrete: capacity does not only buy classification accuracy, it buys *usable probabilistic structure*. A downstream, cost-sensitive decision rule can extract a roughly six-fold improvement in rare-class recall from the MLP's near-continuous probabilities, but can do almost nothing with the tree's 36 piecewise-constant values - independent of, and in addition to, the tree's already-lower classification ceiling.

5.10 The Capacity vs. Interpretability Trade-Off

This cross-model comparison is the central contribution of the report. Using the uniform `Attribution` framework, we compared the interpretable baseline models against the high-capacity sequence model. Table 16 compares each architecture using its most comparable post-hoc method (IG for neural models; native path for the tree).

Table 16: The Capacity vs. Interpretability Trade-Off across architectures. Higher is better on all three metrics; capacity rises down the rows, so the opposing directions of travel are the finding.

Model (Capacity $\rightarrow$)	Macro-F1	Localization	Faithfulness
Decision Tree	0.183	1.000 (tree path)	0.709
Feed-forward MLP	0.288	0.524 (IG)	0.484
Attention RNN	0.345	0.257 (Attn) / 0.386 (IG)	0.193 (Attn) / 0.202 (IG)

The trade-off is monotonic across the entire spectrum. As architectural capacity increases, classification performance steadily improves (macro-F1: 0.18 $\rightarrow$ 0.29 $\rightarrow$ 0.35), while both localization (1.00 $\rightarrow$ 0.52 $\rightarrow$ 0.39) and faithfulness (0.71 $\rightarrow$ 0.48 $\rightarrow$ 0.20) collapse. This dictates two critical conclusions:

(i) Transparent by design does not equal perfectly faithful. The decision tree's localization is 1.00 by construction, as its splits are rigidly bound to the clinical windows. Yet, its native explanation is only 0.71 faithful. Plausibility does not guarantee mathematical reliance.

(ii) Degradation tracks capacity, not explanation methodology. Holding the attribution method strictly constant (Integrated Gradients), the lower-capacity MLP generates maps that are both more plausible (0.524 vs. 0.386) and more faithful (0.484 vs. 0.202) than the sequence model. The loss of legibility is a direct symptom of deep sequence capacity, not an artifact of post-hoc approximation.

To rule out the objection that the tree's native localization of 1.00 is a structural artifact of feeding it clinically-windowed features rather than a measurement, we additionally explain all three models with one identical method: model-agnostic permutation SHAP over the raw 187-sample signal. Each model is read only through its black-box `predict_proba`. For the tree, each perturbed signal is re-featurized on the fly, ensuring SHAP sees it exactly as it sees the neural models: an opaque signal→probability map.

Table 17: Identical-method (SHAP) cross-check across all three architectures. Every score uses the same model-agnostic permutation SHAP over the raw 187-sample signal (zero baseline, black-box `predict_proba`), scored by the same localization and occlusion-faithfulness metrics on the same $n = 30$ DS2 beats.

Model (Capacity →)	**Macro-F1**	**SHAP Localization**	**SHAP Faithfulness**
Decision Tree	0.183	0.866	0.680
Feed-forward MLP	0.288	0.500	0.502
Attention RNN	0.345	0.289	0.241

Table 17 demonstrates that the trade-off survives entirely intact on a single end-to-end scale. SHAP localization falls monotonically (0.866 → 0.500 → 0.289) and SHAP faithfulness degrades identically (0.680 → 0.502 → 0.241) as capacity rises. The tree's measured SHAP localization (0.866) sits below its by-construction native score (1.00) yet remains well above both the 0.385 uniform baseline and the neural models. Furthermore, its SHAP faithfulness (0.680) closely tracks its native-path faithfulness (0.71) - two independent explanation methods in strict agreement.

Table 18: Uncertainty on the explanation metrics. Means with 95% percentile bootstrap confidence intervals (20,000 resamples) over the same $n = 30$ DS2 beats. Per-beat scores were retained for the decision tree and MLP runs only, so intervals are reported for those two architectures.

Model / Method		**Localization**	**Faithfulness**
Decision Tree	native path	1.000 [1.000, 1.000]	0.709 [0.514, 0.873]
Decision Tree	SHAP	0.866 [0.822, 0.909]	0.680 [0.516, 0.842]
Feed-forward MLP	IG	0.524 [0.491, 0.557]	0.484 [0.402, 0.561]
Feed-forward MLP	SHAP	0.500 [0.473, 0.526]	0.502 [0.426, 0.574]

Since all methods are scored on the same 30 beats, the cross-model gaps admit a paired test. On localization the degradation is decisive at every step: the tree exceeds the MLP by $+0.476$ (95% CI $[+0.444, +0.510]$) on the native-path/IG pairing and by $+0.366$ ($[+0.322, +0.410]$) on the identical-method SHAP pairing, both with Wilcoxon signed-rank $p < 0.0001$. Faithfulness is weaker and only directionally supported: the tree-MLP gap is $+0.225$ ($[+0.047, +0.385]$, $p = 0.015$) on the native/IG pairing, but $+0.178$ ($[-0.005, +0.355]$, $p = 0.13$) on the SHAP pairing - that is, the first step of the faithfulness chain is not separable from sampling noise at $n = 30$. We therefore report the localization trend as an established result and the faithfulness trend as directional, and we do not read the within-model differences among attribution methods (e.g. the attention model's 0.193 / 0.202 / 0.241) as a ranking at all, since their spread is well inside the per-beat standard error.

An important methodological caveat applies to the SHAP cross-check: because an isolated waveform beat carries no inter-beat neighbor timing, the tree's on-the-fly featurizer imputes RR_{prev} and RR_{next} at their training means during SHAP perturbation. Consequently, SHAP attributes importance exclusively over the four morphological features (QRS, T-wave, ST-segment). This is the mathematically appropriate signal-level comparison, as RR timing has no location on the waveform itself, and the neural attributions are likewise constrained purely to the signal.

Measurement Caveat (The Input Confound): This trade-off must be analyzed as a *within-model, directional trend* rather than a strict three-way leaderboard, largely due to the sequence model's RR features. First, appending the RR features forces the sequence model's explainers to evaluate the waveform against an out-of-distribution zero-baseline, meaning localization scores are not on a perfectly absolute scale. Second, the input representations and the imbalance strategy themselves confound the capacity comparison: the MLP consumes only the raw waveform and is trained under full inverse-frequency class weighting (macro-F1 0.288), whereas the final sequence model sees the waveform plus RR features under milder square-root weighting (macro-F1 0.345), a weighting choice worth roughly 0.07 macro-F1 to the attention model in the closest matched comparison Section 5.6

affords (0.218 with full weighting versus 0.292 unweighted, both without RR). As demonstrated in the tuning arc (Section 5.6, Config A), the sequence architecture *without* RR features achieves a macro-F1 of 0.292. This means the classification gap between the neural models is substantially driven by the expanded feature set and the weighting difference rather than by pure architectural capacity. Nevertheless, the empirically robust finding remains the directional inversion: pushing a model to higher accuracy - whether via sequence capacity or expanded features - directly collapses explanation transparency. Figure 11 plots that inversion across the three architectures, and Figure 12 shows the per-class recall pattern underneath it.

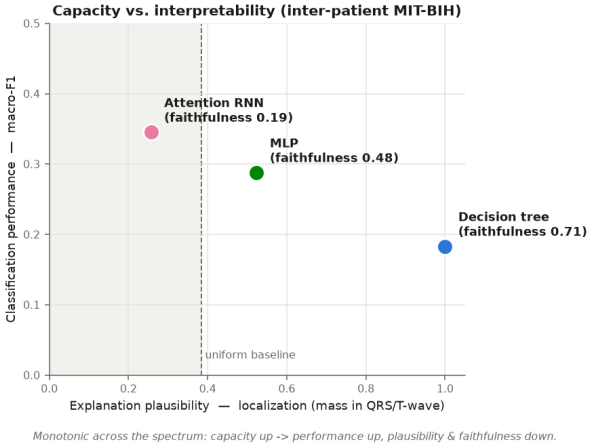


Figure 11: Capacity vs. Interpretability. Classification performance plots inversely to explanation plausibility across all three architectures, illustrating the central trade-off.

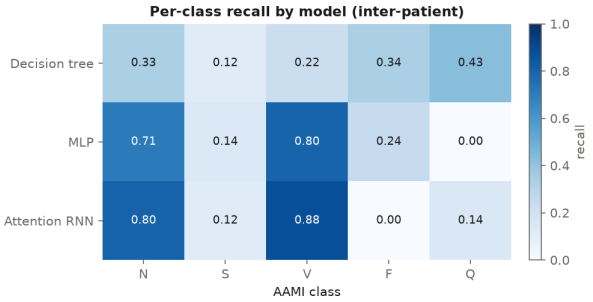


Figure 12: Per-class recall across the model spectrum. Deeper architectures (bottom) increasingly recover the common abnormal V class, while rare classes (S, F, Q) remain uniformly resistant to capacity.

5.11 The Accuracy-Inflation Gap

To demonstrate the critical necessity of the inter-patient evaluation protocol, the identical Attention RNN configuration was re-trained and evaluated under the intra-patient split defined in Section 4.2 (where beats from the same patient can leak across train and test).

Table 19: Accuracy-Inflation Gap: Inter-patient (honest) vs. Intra-patient (leakage).

Metric	**Inter-patient (Honest)**	**Intra-patient (Leakage)**	**Δ**
Accuracy	0.777	0.981	+0.204
Macro-F1	0.345	0.716	**+0.371**
AUROC	0.739	0.946	+0.207
AUPRC	0.375	0.755	+0.380

Under patient leakage (Table 19), macro-F1 **more than doubles** (0.345 → 0.716), and accuracy artificially approaches 98%. Because deep neural models easily memorize patient-specific morphological quirks, the intra-patient numbers are a clinical illusion. This profound inflation justifies the strict enforcement of the De Chazal inter-patient split for all headline metrics, and Figure 13 shows the three headline metrics side by side under both protocols.

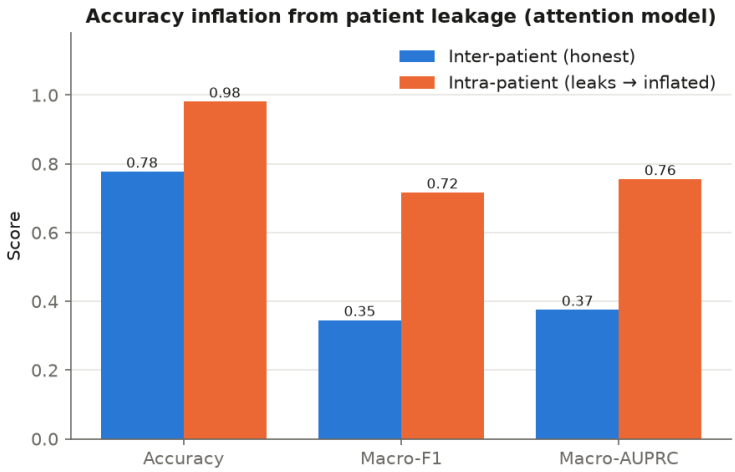


Figure 13: The Accuracy-Inflation Gap. Evaluating the identical model under patient leakage nearly doubles the macro-F1 score, emphasizing the danger of standard split protocols in clinical ML.

The same experiment run on the MLP shows the identical textbook pattern - macro-F1 rises from 0.288 (honest) to 0.544 (leaked), with AUROC and AUPRC inflating by +0.225 and +0.325 respectively. The decision tree, however, is a genuine counter-example (Table 20): its threshold-free ranking metrics (AUROC, AUPRC) inflate as expected under leakage, but its threshold-dependent metrics - accuracy and macro-F1 - actually *drop*.

Table 20: Decision tree accuracy-inflation gap: inter-patient (honest) vs. intra-patient (leakage).

Metric	**Inter-patient (Honest)**	**Intra-patient (Leakage)**	Δ
Accuracy	0.318	0.186	−0.132
Macro-F1	0.183	0.152	−0.031
AUROC	0.671	0.724	+0.053
AUPRC	0.261	0.313	+0.052

The divergence between the ranking metrics and the threshold-dependent metrics is real, but we attribute it to the training-time over-correction flagged in Section 4.3 rather than to anything intrinsic about tree behaviour under leakage. Because the `class_weight` and `sample_weight` corrections multiply, the effective minority-class weighting is roughly squared; once within-patient morphology becomes predictable under leakage, that doubled weighting plausibly pushes the decision boundary further still toward the minority classes, collapsing Normal-class recall enough to drag accuracy and macro-F1 down even as the underlying ranking quality (AUROC/AUPRC) genuinely improves. Confirming this by re-fitting with a single correction was out of scope for this report (Section 6), so this should be read as a flagged anomaly rather than a validated finding. The broader methodological lesson survives either way: "patient leakage always inflates the headline number" is not a uniform law across model families, and the safest guardrail remains the inter-patient split itself, checked directly rather than assumed.

6 Limitations

Severe class imbalance is a property of the data, not of the models. Rare classes (S, F, Q) show low recall under every architecture we tried, inter-patient; Q in particular has only a handful of representative beats in DS2 and is reported for completeness rather than as a reliable estimate. This is why every headline metric in this report is macro-averaged precision/recall/F1 plus AUROC/AUPRC, and never aggregate accuracy, which would be dominated by the 89.5% Normal-class share.

Attention is a candidate explanation, never the last word. Following Jain and Wallace, the attention model's attention weights are always corroborated against an independent post-hoc method (Integrated Gradients, SHAP) and the occlusion-based faithfulness test; the modest (+0.33/+0.34) Spearman agreement in Section 5.8 is itself evidence for this caution, not against it. The attention head is also class-agnostic - a single map is used to explain every class's prediction on a given beat, which necessarily coarsens what the explanation can say about class-specific reasoning.

The RR out-of-distribution artifact bounds how absolute the central trade-off can be read. Because the attention model's RR features are appended at the classifier head rather than fed through the recurrence, post-hoc explainers evaluating the waveform

in isolation must hold RR at an out-of-distribution zero baseline. This is why Section 5.10's headline trade-off is presented as a directional, within-model trend rather than a single strictly-comparable three-way leaderboard.

The decision tree's probabilities are leaf-coarse by construction. A depth-6 tree's `predict_proba` takes only 36 distinct values across the entire 49,698-beat test set (distributed across its 57 leaves) - a granularity limitation entirely independent of accuracy, and the direct explanation for why cost-sensitive re-thresholding can do so little for the tree relative to the MLP (Section 5.9).

The tree's double class-imbalance correction is a probable methodological error, not a design choice. The `class_weight` and `sample_weight` corrections are applied simultaneously during training, and because scikit-learn multiplies the two, the minority classes are effectively up-weighted twice over. We regard this as the most likely cause of both the tree's collapsed Normal-class recall and its inflation gap running backwards from the other two models (Section 5.11). Re-fitting with a single correction would confirm this and is the first change we would make in a revision; it was out of scope here, so all tree numbers should be read as as-trained under an over-corrected weighting.

Per-beat amplitude normalization discards absolute scale. Beats are min-max normalized individually to $[0, 1]$, so amplitude-derived features (QRS amplitude, T-wave amplitude) are relative to each beat's own dynamic range rather than absolute millivolts; this is standard in the ECG literature but means the tree's amplitude-based splits should be read as relative thresholds.

Neither the neural models nor the tree is strictly reproducible. The MLP's dropout and Adam optimizer introduce run-to-run non-determinism across machines; the decision tree is far more stable but its `random_state` is left unset, so ties between equally-scoring candidate splits are broken non-deterministically rather than under a fixed seed. All MLP numbers reported in this document are drawn from a single canonical trained checkpoint, re-evaluated identically for every metric, explanation, and decision-theoretic analysis, to ensure internal consistency across Sections 5.3–5.4 and 5.9–5.11.

7 Conclusion

Across three architectures spanning the transparency spectrum - a depth-bounded decision tree over clinically-grounded features, a feed-forward network over the raw waveform, and a bidirectional attention-based recurrent model - this report finds a consistent, directional trade-off: primary classification performance generally rises with model capacity (macro-F1 0.183 $\rightarrow$ 0.288 $\rightarrow$ 0.345, though the MLP marginally peaks on threshold-free AUROC at 0.747 vs. 0.739) while both plausibility (localization 0.87 $\rightarrow$ 0.50 $\rightarrow$ 0.29) and faithfulness (0.68 $\rightarrow$ 0.50 $\rightarrow$ 0.24, directionally - see Table 18) of the model's explanations fall. This trend holds even when the explanation methodology is held fixed - the mid-capacity MLP's Integrated Gradients maps are more plausible and more faithful than the high-capacity attention model's, isolating capacity itself, rather than the choice of explainer, as the driver of the degradation.

A second, orthogonal finding concerns decision theory rather than explanation: cost-sensitive re-thresholding can lift a model's rare-class recall dramatically, but only if the model's probability outputs are fine-grained enough to be moved. The MLP's near-continuous probabilities let a single cost matrix raise S-class recall roughly six-fold (0.14 $\rightarrow$ 0.84); the tree's 36 piecewise-constant leaf probabilities give the identical cost matrix almost nothing to work with. Capacity, in other words, governs not only what a model can classify, but how much a downstream clinical decision rule can subsequently do with its output.

A third finding is methodological rather than architectural: the patient-disjoint split is not merely good practice but empirically load-bearing, and unevenly so. Patient leakage roughly doubles the attention model's and the MLP's macro-F1, the textbook inflation pattern - but the decision tree's threshold-dependent metrics actually *fall* under the same leakage, most likely an artifact of its aggressive double class-imbalance correction interacting with the intra-patient split. No single story about leakage holds uniformly across model families, which is itself an argument for verifying the patient-disjoint guardrail directly (as this project's test suite does) rather than assuming its effects.

Taken together, these results argue that a model's raw predictive accuracy is an incomplete - and in a clinical decision-support context, potentially misleading - summary of its value. A uniform attribution interface that scores every architecture's explanation on the same plausibility and faithfulness axes turns "is this model trustworthy" from a qualitative impression into a quantitative, comparable measurement, and this report's central result is that, on ECG arrhythmia classification, higher diagnostic capacity is purchased directly at the expense of that trust.

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